Reg. No.															

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024
Fifth Semester

21CSC302J – COMPUTER NETWORKS
(For the candidates admitted from the academic year 2022-2023 onwards)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks) Marks BL CO
Answer **ALL** Questions

1. LAN interconnects hosts; a WAN interconnects connecting _____ 1 1 1
   (A) Networking devices (B) LANs
   (C) PCs (D) PANs

2. A_____ is always available between the two end systems which becomes active or inactive based on the switch 1 1 1
   (A) Packet switched network. (B) point-to-point WANs
   (C) circuit-switched network (D) Networking devices

3. The process-to-process delivery of the entire message is the responsibility of the _____ layer.
   (A) Network (B) Transport
   (C) Application (D) Physical

4. Which of the following primarily uses guided media? 1 1 1
   (A) Cellular telephone system (B) Local telephone system
   (C) Satellite communication (D) Radio Broadcasting

5. _____ is the multiport repeater. 1 2 2
   (A) Hub (B) Bridge
   (C) Switch (D) Router

6. Given the IP address 201.14.78.65 and the subnet mask 255.255.255.224 what is the subnet address? 1 2 2
   (A) 201.14.78.32 (B) 201.14.78.65
   (C) 201.14.78.64 (D) 201.14.78.12

7. Class _____ has the greatest number of hosts per given network address 1 2 2
   (A) A (B) B
   (C) C (D) D

8. On a network that uses NAT (Network Address Translation), the _____ has translation table. 1 1 2
   (A) Switch (B) Bridge
   (C) Server (D) Router

9. In distance vector routing each router receives information directly from _____. 1 1 3

(A) Every router on the network (B) Every router less than two units away
(C) A table stored by the network hosts (D) Its neighbours only

10. Router B receives an update from router A that indicates Netl is two hops away. The next update from A says Netl is five hops away. What value is entered in B's routing table for Netl? Assume the basic RIP is being used 1 2 3

(A) 2 (B) 3
(C) 6 (D) 7

11. Dijkstra's algorithm is used to _____. 1 1 3

(A) Create LSAs (B) Flood an internet with information
(C) Calculate the routing tables (D) Create a link state database

12. _____ is not a multicasting application 1 2 3

(A) Teleconferencing (B) Distance learning
(C) Information dissemination (D) Train announcements

13. In the _____ Random access method, there is no collision 1 1 4

(A) Aloha (B) Csma/ca
(C) Csma/cd (D) Token passing

14. Flow control is needed to prevent _____. 1 1 4

(A) Bit errors (B) Overflow of the sender buffer
(C) Overflow of the receiver buffer (D) Collision between sender and receiver

15. For a sliding window of size n-1 (n sequence numbers), there can be maximum of _____ frames sent but unacknowledged 1 1 4

(A) 0 (B) n-1
(C) n (D) n+1

16. HDLC is an acronym for _____. 1 1 4

(A) High-duplex line communication (B) High-level data link control
(C) Half-duplex digital link (D) Host double-level circuit combination

17. UDP needs the _____ address to deliver the user datagram to the correct application program 1 1 5

(A) Application (B) Max
(C) IP (D) Port

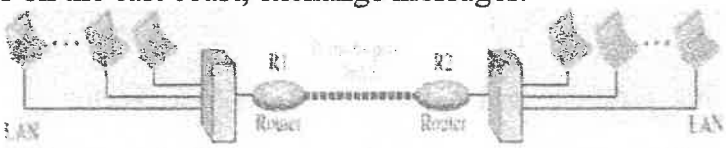
18. The _____ defines the server program 1 1 5

(A) Ephemeral Port number (B) IP Address
(C) Well known port number (D) MAC address

19.	To prevent silly window syndrome created by a receiver that processes data at a very slow rate _____ can be used (A) Clark's solution (B) Nagle's algorithm (C) Delayed acknowledgement (D) Zero timer	1	1	5
20.	The host with the domain name srm.sco.ktr.univ is on the _____ level of the DNS hierarchical tree (The root is level 1) (A) 4th (B) 5th (C) 6th (D) 7th	1	2	5

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

		Marks	BL	CO
21. a.	Using the internet in Figure shown , show the layers of the TCP/IP protocol suite and the flow of data when two hosts, one on the west coast and the other on the east coast, exchange messages.  West coast office East coast office	8	2	1
	(OR)			
b.i.	Why does a circuit-switched network need end-to-end addressing during the setup and teardown phases? Why are no addresses needed during the data transfer phase for this type of network?	4	2	1
ii.	Why does a packet switching network need only end-to-end addressing during the data transfer phase, but no addressing during the setup and teardown phases?	4	2	1
22. a.	In contemporary routing and addressing, the notation commonly used is called classless interdomain routing or CIDR. With CIDR, the number of bits in the mask is indicated in the following fashion: 192.168.100.0/24. This corresponds to a mask of 255.255.255.0. If this example would provide for 256 host addresses on the network, how many addresses are provided with the following? (i) 192.168.100.0/23 (ii) 192.168.100.0/25	8	2	2
	(OR)			
b.	A switch uses a filtering table; a router uses a routing table. Can you explain the difference considering their functionalities and features?	8	2	2
23. a.	Brief about the process of classful and classless subnetting with examples.	8	1	3
	(OR)			
b.	Explain the EIGRP Protocol with required diagrams .	8	1	3
24. a.	List some strategies in CSMA/CA that are used to avoid collision. 8 1 3			
	(OR)			
b.	Explain Error Detection strategies in Data Link Layer.	8	1	3

25. a. Compare the working of TCP and UDP. 8 1 5

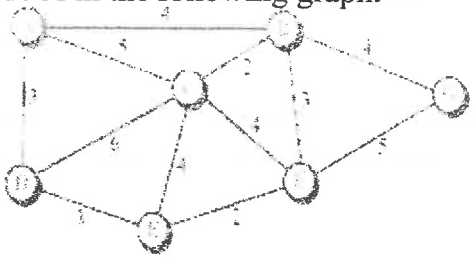
(OR)

b. Write about Application layer protocols : HTTP and DNS. 8 1 5

PART – C (1 × 15 = 15 Marks) Marks BL CO

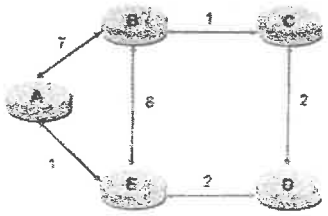
Answer **ANY ONE** Question

26. Use Dijkstra's algorithm to find the shortest path tree and the forwarding table for node A in the following graph. 15 1 3

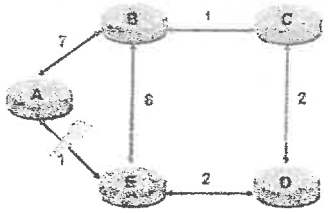


Write the advantages of this protocol in comparison to distance vector routing.

27. Find the final convergence table for the following network using Bellman-ford algorithm. 15 1 3



Assume the link between A and E is broken as shown below. Explain the count to infinity problem that occurs due this broken link asnd also prove split horizon with poison reverse is the solution.



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